ET HACKATHON · AI-DRIVEN ENERGY SUPPLY-CHAIN RESILIENCE

EnergyGuardian AI

An AI-powered decision-support platform for energy supply-chain resilience — built around India and the Strait of Hormuz.

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EnergyGuardian transforms publicly available geopolitical, logistics, and market signals into transparent risk assessments, scenario analyses, and evidence-backed recommendations that help analysts respond more quickly to emerging disruptions. It is explicitly a decision-support system: every output is explainable and traceable to a documented assumption, and it augments human decision-makers rather than replacing them.

1 The problem

India imports about **88% of the crude oil it burns**, and close to **half of that** arrives through a single 33-km-wide chokepoint — the Strait of Hormuz. When that lane is threatened, refiners have hours to react and the country holds roughly a **nine-and-a-half-day** strategic cushion. Today the response to a shock is largely manual: analysts read the news, debate severity, and reroute cargoes days later.

88%	~45%	~9.5 d	+47 d
crude imported	of imports via Hormuz	strategic reserve cover	slower recovery without a response layer*

*McKinsey estimate for economies lacking a response-intelligence capability. Every figure above is sourced in the in-app Assumptions Register.

The gap is not data — public signals exist — but a layer that **continuously analyzes those signals**, connects them to physical supply dependencies, and turns them into an explainable, end-to-end response: from signal to recommendation.

2 Core innovation — the compound-risk model

Traditional monitoring evaluates disruption indicators independently, one threshold at a time. EnergyGuardian introduces a **compound-risk model** that explicitly captures nonlinear interactions between market stress, freight disruption, and geopolitical events. Rather than waiting for a single threshold crossing, the model identifies combinations of individually weak signals whose *interaction* indicates elevated systemic risk.

For each day, robust z-score anomalies (median/MAD over a 14-day window, clipped to [-3, 3]) are combined:

$$\mathrm{Risk}(t) = \Sigma\, w_i \cdot a_i(t) \;+\; \lambda\, \Sigma\, a_i^+(t) \cdot a_j^+(t)$$

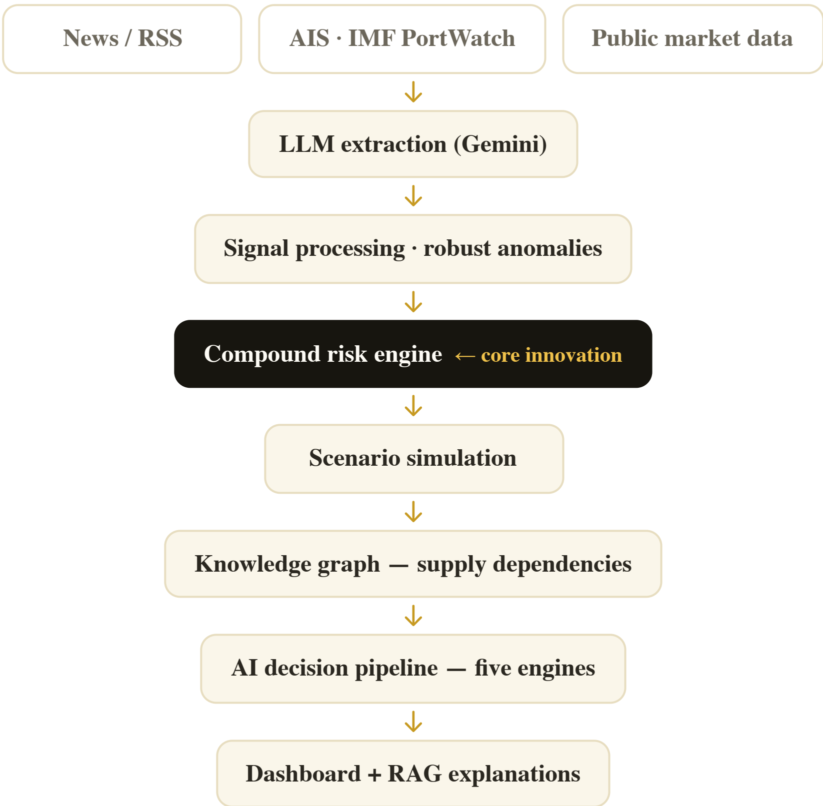
A signed linear term (each signal's own contribution) plus a co-elevation interaction term that fires only when signals are *simultaneously* elevated. The interaction term is the innovation.

Every recommendation the system produces is accompanied by:

- the **contributing signals** and their individual weights;
- the **documented assumptions** behind every parameter;
- **explainable reasoning** — a plain-language account of what drove the score; and
- a **comparison against a traditional per-signal baseline**.

3 System architecture

A thin Flask API serves a display-only React dashboard; all computation lives in the backend. Data flows in one direction, signal to recommendation:



4 AI workflow

Six AI/analytics capabilities each do a specific job — not decoration. Where a stage is deterministic, it is described as such; the one generative-LLM step is news extraction.

COMPONENT	PURPOSE
Gemini LLM extraction	Convert unstructured news (live or pasted) into structured disruption events — corridor, severity, event type, confidence. Transparent keyword fallback offline.
Compound risk model	Evaluate nonlinear interactions between disruption indicators to surface systemic, multi-signal risk.
Predictive scenario simulation	Project the cascade — Brent premium, import gap, refinery run-rate, GDP trajectory — from a given risk posture.
AI decision pipeline	Coordinate five deterministic engines (signal → scenario → procurement → reserve → analogue) over a shared context to generate and explain recommendations; every step's input, reasoning and output is inspectable.
Knowledge graph	Model supplier ↔ corridor ↔ grade ↔ refinery dependencies and trace disruption impact by graph traversal.
Retrieval-augmented generation (RAG)	Provide grounded, cited explanations from a documented energy-security corpus.

Geospatial intelligence (corridor/chokepoint mapping with AIS-derived tanker traffic) underlies the scenario and knowledge-graph layers. The five-stage pipeline uses the orchestration pattern behind agentic systems, but the stages are deterministic engines — named honestly rather than described as autonomous "AI agents."

5 Validation against historical events

The compound model was evaluated on the 2023–24 Red Sea / Hormuz period against documented disruption events, not on synthetic data.

Dataset

- Red Sea & Hormuz corridor event log (public reporting)
- Reuters / news headlines (Google News RSS)
- AIS-derived chokepoint traffic (IMF PortWatch)
- Public market data (Brent) and documented freight prints

Method

- **Baseline:** independent per-signal threshold model
- **Metrics:** true positives, false positives, lead time
- **Replay:** historical window, Oct 2023 – Mar 2024
- 10 documented events (severity ≥ 4) as ground truth

METHOD	TRUE POSITIVES	FALSE POSITIVES	ALERT DAYS
Per-signal baseline	1	3	4
Compound model	**3**	**2**	5

The compound model detected 3× the events with fewer false alarms at comparable alert volume. Reported honestly: lead time to the systemic onset is identical for both, and both miss isolated single incidents — the model is built to surface clustered, systemic disruption, and the document says so.

6 Business impact & scalability

Primary users

- Indian Oil Corporation (IOC)
- Bharat Petroleum (BPCL)
- Hindustan Petroleum (HPCL)
- Strategic Petroleum Reserve operators (ISPRL)
- Ministry of Petroleum & Natural Gas
- Energy trading desks

Impact & scale

- **Impact:** compressing the analyse-and-respond loop from days to minutes directly attacks the "+47 days slower recovery" penalty. On India's import volumes, shaving even a few days off a reroute decision during a Brent spike is measured in hundreds of millions of dollars and fractions of a GDP point.
- **Scalability:** the architecture is corridor- and country-agnostic. The same engine extends to LNG, to other importers (Japan, Korea, the EU), and to other chokepoints (Malacca, Bab-el-Mandeb, the Bosphorus) by swapping the data layer — the model and UI are unchanged.
- **Cost:** it runs on free and public data sources and a free LLM tier, so it is deployable today at near-zero marginal cost.

7 Limitations

- The current implementation evaluates historical and simulated scenarios; operational deployment would require integration with enterprise logistics systems.
- Some scenario parameters are assumption-based and are explicitly documented in the Assumptions Register (event severity is an assigned ordinal; the freight signal is sparse, with forward-filling stated as a modelling choice, not fabricated data).
- Procurement recommendations are intended as decision support, not automated execution.
- Model performance has been demonstrated on public historical events and should be further validated with proprietary operational datasets before production deployment.
- The only generative-LLM use is extracting and rewording; it never computes a risk number.

8 Conclusion

EnergyGuardian demonstrates how explainable AI can improve energy supply-chain resilience by transforming heterogeneous public information into transparent operational intelligence. Rather than replacing human decision-makers, the platform enables analysts to:

- identify emerging disruption patterns,
- evaluate cascading impacts,
- explore alternative response scenarios, and
- understand the reasoning behind every recommendation.

The architecture is modular, explainable, and readily extensible to additional countries, commodities, and transport corridors, making it a practical foundation for next-generation energy-resilience systems.